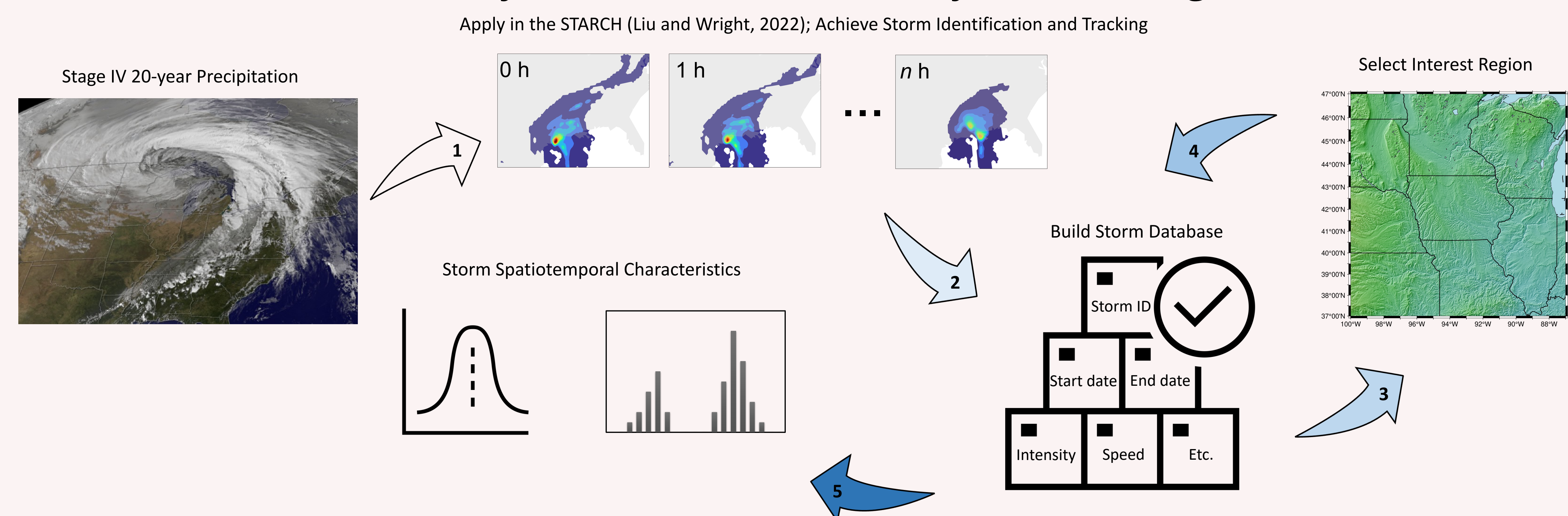


Motivation

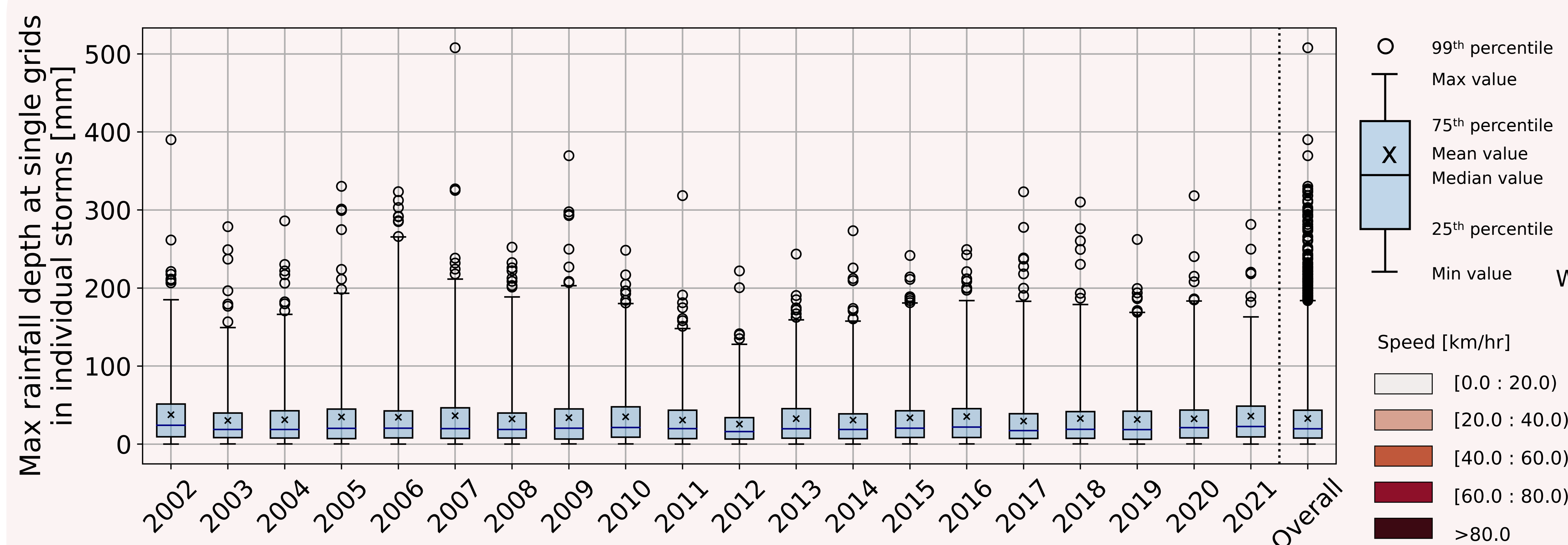
The emergence of multidecadal high-resolution gridded precipitation datasets provides opportunities to advance understanding and modeling of extreme storms, including for rainfall/flood frequency analysis and probable maximum precipitation/flood estimation. By studying the intra-storm rain intensity variability, the precipitation characteristics and large-scale synoptic characteristics associated is documented for heavy-to-extreme precipitation in the upper Midwestern United States.

Methodology

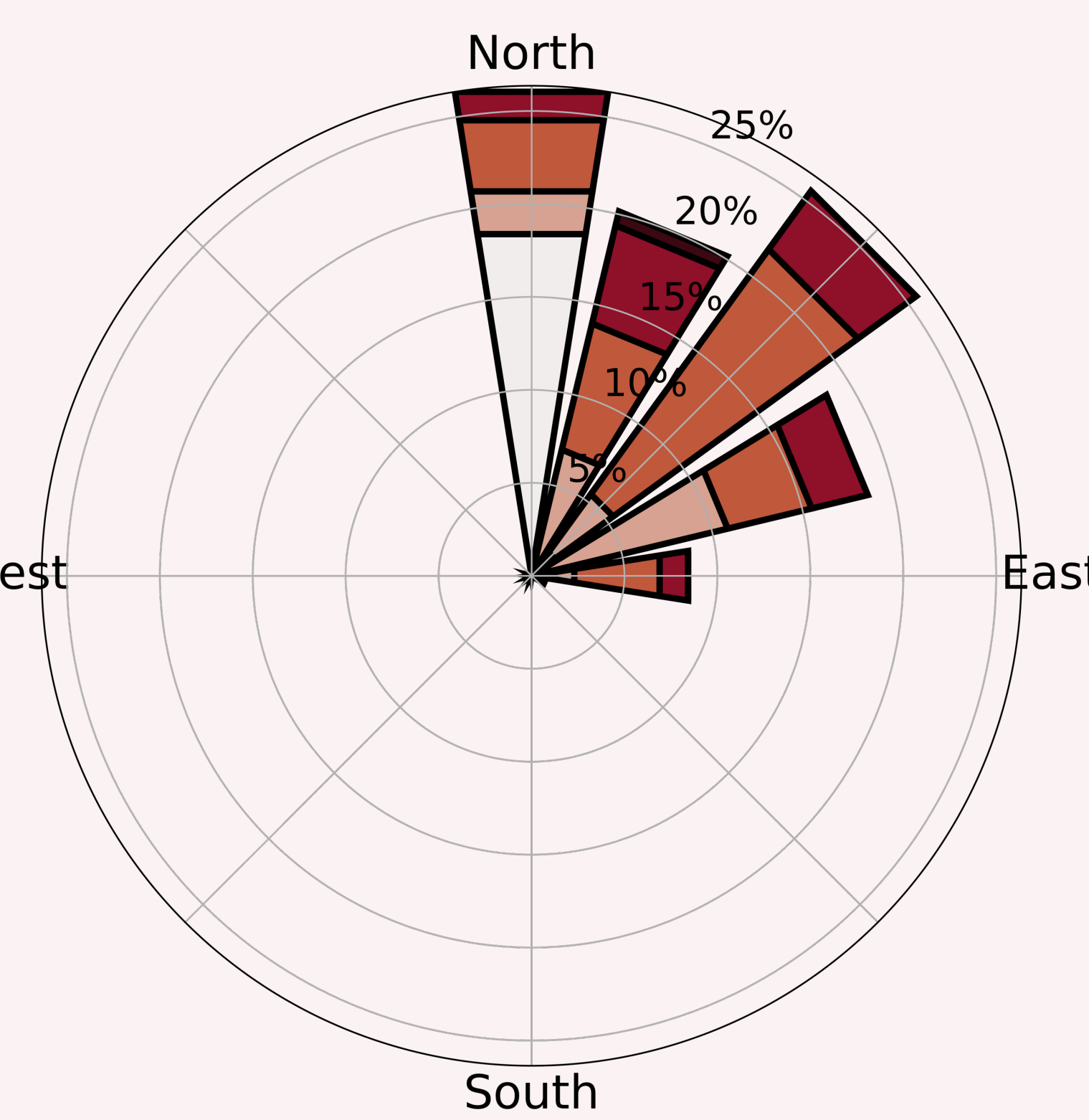
The 4 km × 4 km hourly NCEP Stage IV precipitation analysis, based primarily on gage-corrected weather radar, provides accurate precipitation estimates east of the Rocky Mountains. In this study, we use Stage IV database together with the Storm Tracking And Regional Characterization algorithm (STARCH) to study the characteristics of heavy precipitation events across all seasons in the upper Midwestern United States for the 20-year period from 2002 to 2021.



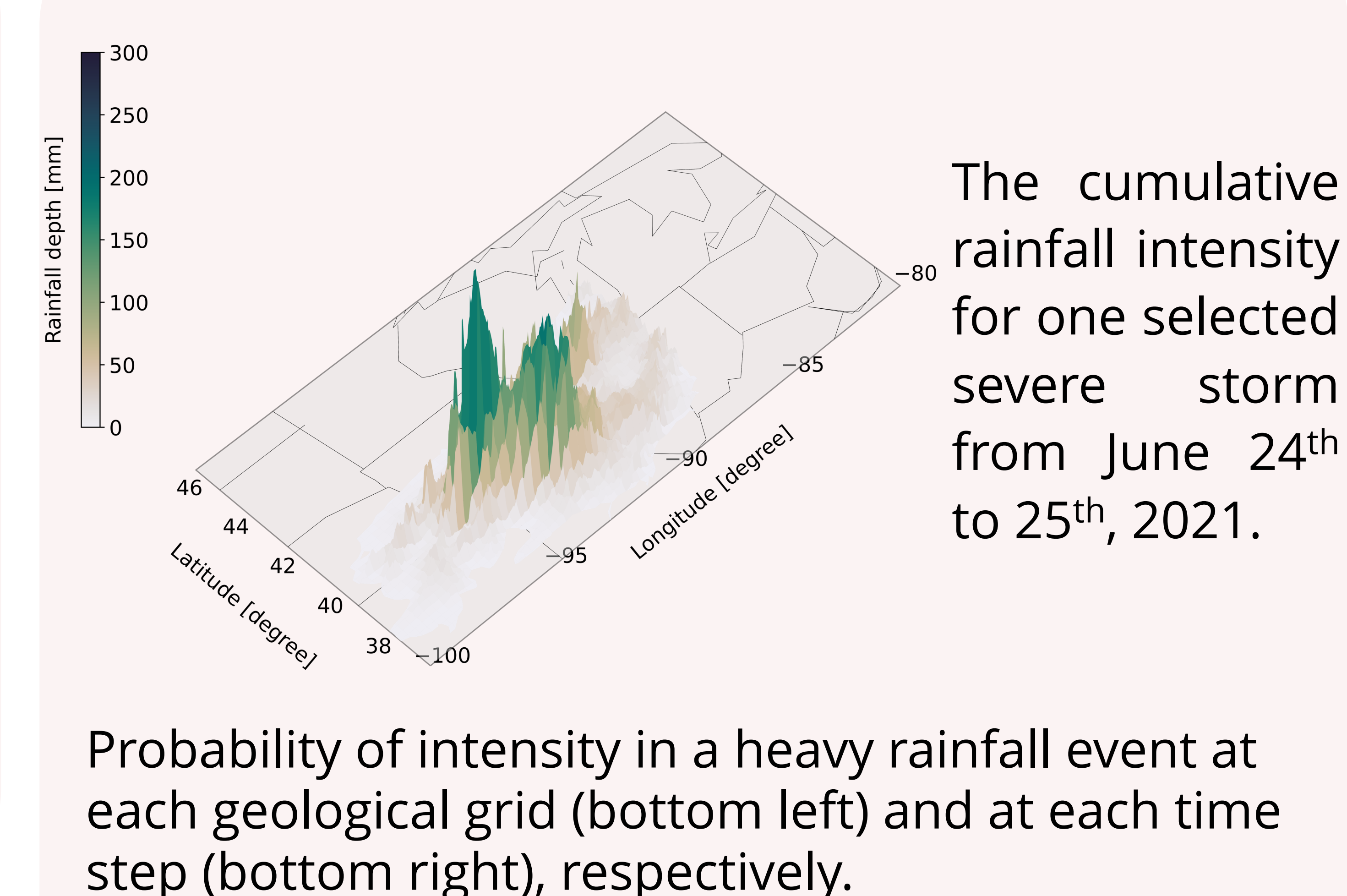
Heavy Precipitation



(left) Maximum intensities of summer storms accumulated through time series. (right) A rose diagram presenting the distribution of 99th percentile heavy storms motion vector (moving speed and direction).



Case Study



Probability of intensity in a heavy rainfall event at each geological grid (bottom left) and at each time step (bottom right), respectively.

Takeaway

For heavy-to-extreme precipitation, the distribution of intra-storm varies. Temporal and spatial distributions change at different locations through time.

Next Steps

Analyze spatiotemporal characteristics for extremely convective storms through four seasons and propose detailed suggestions for stochastic rainfall generators. Study large-scale synoptic characteristics.

